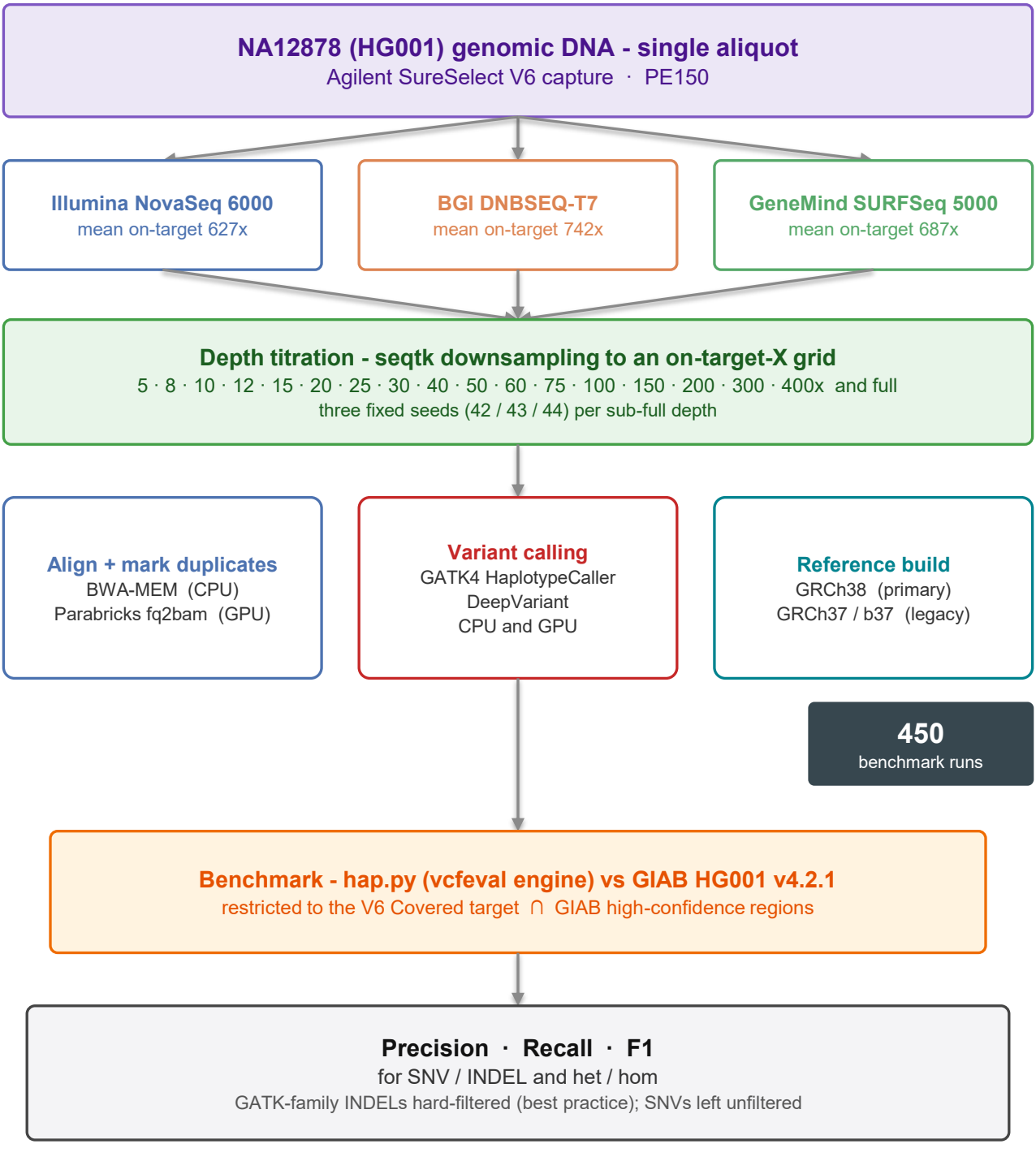

NA12878 (HG001) genomic DNA - single aliquot
Agilent SureSelect V6 capture · PE150
Illumina NovaSeq 6000
mean on-target 627x
BGI DNBSEQ-T7
mean on-target 742x
GeneMind SURFSeq 5000
mean on-target 687x
Depth titration - seqtk downsampling to an on-target-X grid
5 · 8 · 10 · 12 · 15 · 20 · 25 · 30 · 40 · 50 · 60 · 75 · 100 · 150 · 200 · 300 · 400x and full
three fixed seeds (42 / 43 / 44) per sub-full depth
Align + mark duplicates
BWA-MEM (CPU)
Parabricks fq2bam (GPU)
Variant calling
GATK4 HaplotypeCaller
DeepVariant
CPU and GPU
Reference build
GRCh38 (primary)
GRCh37 / b37 (legacy)
450
benchmark runs
Benchmark - hap.py (vcfeval engine) vs GIAB HG001 v4.2.1
restricted to the V6 Covered target ∩ GIAB high-confidence regions
Precision · Recall · F1
for SNV / INDEL and het / hom
GATK-family INDELs hard-filtered (best practice); SNVs left unfiltered